BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. A person ascending on the building and reached at 5th floor,then the value of "g" for this situation will be:

(A) Increasing
(B) Decreasing
(C) Remained same
(D) Zero

Q2. An object having weight 120 N lifted through an height of 5 m , work done on object is:

(A) 24 J
(B) 240 J
(C) 60 J
(D) 600 J

Q3. SI unit of energy is:

(A) newton
(B) joule
(C) watt
(D) pascal

Q4. The air surrounds us exerts force in:

(A) Upward direction only
(B) Downward direction only
(C) Two direction
(D) All direction

Q5. If ΔQ is the change in heat and ΔT is the change in temperature, the heat capacity is:

(A) ΔQ x ΔT
(B) ΔQ/ ΔT
(C) mΔQ x [illegible in source scan]
(D) mΔQ[illegible in source scan]

Q6. The mechanism of transfer of heat in solids is due to:

(A) Conduction only
(B) Convection only
(C) Conduction and Convection
(D) Conduction and radiation

Q7. The pair of derived physical quantities is:

(A) Current, Temperature
(B) Density, Mass
(C) Length, Mass
(D) Area, Volume

Q8. A pigeon flies 20km from nest and moves back to nest, his displacement is:

(A) 0 km
(B) 10 km
(C) 20 km
(D) 40 km

Q9. The property of body to resist the change in its state of rest or motion is called:

(A) Force
(B) Momentum
(C) Inertia
(D) Im[illegible in source scan]

Q10. The force of 10N acting on body of mass 1kg moving along circular path with velocity 2[illegible in source scan] of circular path is:

(A) 2m
(B) 20m
(C) 4m
(D) [illegible in source scan]

Q11. Fx=4N, Fy=3N, F=?

(A) 3N
(B) 4N
(C) 5N

Q12. When force is applied at the pivot point torque produced will be:

(A) Maximum
(B) Minimum
(C) Zero
(D) Ne[illegible in source scan]

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Define the following branches of Physics? A. Heat & Thermodynamics B. Cosmology & Astrophysics C. Bio Physics & Medical Physics D. Electricity & Magnetism

(ii) Define scalar quantities.Give two examples along with symbols?

(iii) If a force 80N acts on a body its momentum changes to 40kg m/s.What is time for which the force acts on the body?

(iv) What is centripetal acceleration? Write its mathematical form.

(v) State graphical method used to find resultant of forces, when forces are neither parallel nor anti parallel!?.

(vi) Define equilibrium?Briefly explain dynamic equilibrium.

(vii) Define Solar energy. Describe the process by which solar energy converts into electricity.

(viii) Calculate the power of a pump which can lift 200kg of water through a height of 6m in 10s.

(ix) Explain Young's Modulus.

(x) Define evaporation. How presence of water vapours in air effects the rate of evaporation.

(xi) Explain the term "sea breeze" and "land breeze" in connection with convection of heat.

SECTION "C"

Maximum Marks: 21

> Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Use velocity-time graph to prove that. $S=vit + ½ at^2$

(b) [3] A ball is dropped from a height of the building reaches to ground in 8.2s Find height of the building, and velocity of the ball,when it reaches the ground.

QIV.

(a) [4] Derive equation to measure the velocity of any planet moving in circular path.

(b) [3] The mass of sun is 1.99×10^{30}kg and the radius of the Earth's orbit around the sun is 1.5×10^{11}m .Calculate the orbital speed of the earth.

QV.

(a) [4] What is Archimedes Principle, Write its mathematical form.

(b) [3] The deepest point in the ocean is 11km below sea level, deeper than Mount Everest is tall, what is the pressure in atmospheres at this depth?

QVI.

(a) [4] Explain linear thermal expansion of solids. On which factors does it depend.

(b) [3] A railway line made of iron is 1200km long and is laid at 25°C.By how much will it expand in summer when temperature rises to 40° C.